

Day 4: Component-Specific Troubleshooting Practical: Identify and fix issues in faulty components (RAM, GPU, storage). Task: Create a checklist for diagnosing hardware issues.

Tasks:-

1. Open your desktop or laptop, locate the RAM slots, and remove and re-insert the RAM module to simulate reseating — then document each step you took and any issues you observed during the process.

Ans.:-

RAM Reseating Procedure Report:-

Objective

To safely remove and reinsert a RAM module in a desktop/laptop to simulate RAM reseating and verify proper installation.

Tools Required

- Phillips screwdriver (if required)
- Anti-static wrist strap (recommended)
- Clean, non-conductive work surface

Steps Performed

Step 1: Shut Down the Computer

- Shut down the operating system.
- Turned off the power supply (desktop) or disconnected the charger (laptop).
- Removed the battery if it was removable.

Step 2: Disconnect Power

- Unplugged the power cable and all external devices.
- Pressed the power button for about 10 seconds to discharge any remaining electrical power.

Step 3: Open the Computer

- Removed the side panel (desktop) or back cover (laptop) using a screwdriver.
- Located the RAM slots on the motherboard.

Step 4: Remove the RAM Module

- Pushed the retaining clips on both sides of the RAM slot outward.
- The RAM module popped up slightly.
- Carefully removed the RAM by holding it at the edges without touching the gold contacts.

Step 5: Inspect the RAM

- Checked for dust, dirt, or visible damage.
- Verified that the gold contacts were clean.

Step 6: Reinsert the RAM

- Aligned the notch on the RAM module with the key in the RAM slot.
- Pressed down firmly and evenly until both retaining clips locked into place automatically.

Step 7: Reassemble the Computer

- Reinstalled the side panel/back cover.
- Reconnected the power cable and peripherals.

Step 8: Test the System

- Powered on the computer.
- Confirmed that the system booted normally.
- Verified that the installed RAM was correctly detected in BIOS or the operating system.

Observations

- RAM module was removed and reinserted successfully.
- Retaining clips locked securely after installation.
- No visible physical damage was observed.
- Computer started normally after reseating.
- BIOS/Windows detected the correct amount of installed RAM.

Issue	Possible Cause	Solution
Computer does not boot	RAM not fully seated	Remove and reinstall the RAM correctly
Continuous beeping	Loose or faulty RAM	Reseat or test another RAM module
Less RAM detected	Module not properly inserted	Reinstall and verify both clips are locked
Frequent crashes	Dirty contacts or defective RAM	Clean contacts carefully or replace the module

2. Create a troubleshooting checklist for diagnosing a faulty GPU in a gaming PC, including at least 5 steps from initial symptom identification to confirming the fix.
Hint: Think about display issues, driver checks, and physical connections.

Ans.:-

Step	Troubleshooting Action	Purpose	Expected Result
1. Identify the Symptoms	Observe the problem, such as no display, screen flickering, graphical artifacts, game crashes, or poor FPS.	Determine whether the issue is related to the GPU.	Symptoms are clearly identified for further diagnosis.
2. Check Physical Connections	Turn off the PC, unplug the power, ensure the GPU is fully seated in the PCIe slot, and verify all PCIe power connectors are securely connected. Check the monitor cable (HDMI/DisplayPort).	Rule out loose hardware connections.	GPU and display cables are securely connected.

3. Verify GPU Drivers	Open Device Manager and check for driver errors. Update or reinstall the latest graphics driver from the GPU manufacturer's website (NVIDIA, AMD, or Intel).	Fix software or driver-related issues.	GPU is recognized correctly with no driver errors.
4. Monitor GPU Temperature and Performance	Use monitoring software such as GPU-Z, MSI Afterburner, or HWMonitor to check GPU temperature and usage while gaming.	Detect overheating or abnormal performance.	GPU temperature remains within the normal range (typically 30–50°C idle and 60–85°C under load).
5. Test the Hardware	Run a GPU stress test or benchmark (e.g., FurMark or 3DMark). If available, test the GPU in another PC or install a known-good GPU in the current system.	Confirm whether the GPU hardware is faulty.	The GPU passes the test without crashes, artifacts, or overheating.
6. Confirm the Fix	Restart the PC, launch games, and verify stable graphics, normal temperatures, and expected performance.	Ensure the problem has been resolved.	Games run normally with no display issues or crashes.

3. Simulate a storage issue by unplugging and replugging a SATA or NVMe drive (SSD/HDD) on a test system, then describe the error messages or BIOS indicators you see before and after reconnecting.

Ans.:-

Steps Performed :-

1. Shut down the computer completely.
2. Turned off the power supply and unplugged the power cable.
3. Opened the computer case.
4. Disconnected the storage drive:
 - SATA drive: Removed the SATA data cable and/or SATA power cable.
 - NVMe SSD: Removed the retaining screw and carefully lifted the SSD from the M.2 slot.
5. Closed the case (or left it open for testing) and powered on the system.
6. Observed the BIOS and startup messages.
7. Powered the system off again.
8. Reconnected the SATA or NVMe drive securely.
9. Restarted the computer and verified that the drive was detected.

Typical Observations :-

Stage	BIOS/Startup Indicator	Explanation
Before reconnecting (drive disconnected)	No Boot Device Found, Boot Device Not Detected, or No Bootable Device	The BIOS cannot find the operating system because the storage drive is missing.
	BIOS Storage Information: SATA Port Empty or NVMe: Not Detected	The disconnected drive is not listed among installed storage devices.

Stage	BIOS/Startup Indicator	Explanation
After reconnecting (drive connected)	Drive appears in BIOS storage list	The BIOS detects the SSD/HDD successfully.
	Windows/Linux starts normally	The operating system loads because the boot drive is available again.

Possible Error Messages:-

- No Boot Device Found
- Boot Device Not Detected
- Operating System Not Found
- Reboot and Select Proper Boot Device
- No Bootable Device

4. Given a scenario where a PC randomly restarts, list which component (RAM, GPU, storage) you would check first and explain your reasoning, then outline the specific diagnostic steps you would perform for that component.

Ans.:-

Component to Check First: RAM (Memory)

Reasoning

RAM is the first component I would check because faulty or improperly seated memory is one of the most common causes of random restarts, system crashes, and Blue Screen of Death (BSOD) errors. If the RAM is unstable or has bad memory cells, the operating system may restart unexpectedly to recover from memory errors.

Diagnostic Steps for RAM

1. Check Physical Installation

- Turn off the PC and unplug the power.

- Open the case and ensure the RAM modules are fully seated in their slots.
- If necessary, remove and reinsert the RAM modules.

Expected Result: RAM is securely locked in place.

2. Inspect the RAM

- Check for dust, corrosion, or visible damage on the RAM sticks and slots.
- Clean dust using compressed air if needed.

Expected Result: RAM and slots are clean and undamaged.

3. Run Windows Memory Diagnostic

- Press Windows + R.
- Type mdsched.exe and press Enter.
- Select Restart now and check for problems.
- Allow the test to complete.

Expected Result: No memory errors are reported.

4. Test RAM Modules Individually

- If multiple RAM sticks are installed, remove all but one.
- Boot the PC and test each module one at a time.
- Repeat using different RAM slots if necessary.

Expected Result: Identify whether a specific RAM stick or slot is faulty.

5. Monitor System Stability

- After reseating or replacing the RAM, use the computer normally or run stress tests for a while.
- Check whether random restarts continue.

Expected Result: The system remains stable without unexpected restarts.

If the RAM Is Not the Problem:-

Continue troubleshooting other components:

- **GPU:** Check temperatures, update/reinstall graphics drivers, verify PCIe power connections, and stress-test the graphics card.
- **Storage (SSD/HDD):** Check SMART health using tools like CrystalDiskInfo, inspect SATA/NVMe connections, and scan the drive for errors.